Software Requirements Specification (SRS)

IEEE 830-style specification, 11 FRs, 6 measurable NFRs, MoSCoW priorities and traceability matrix

1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for **ClubSphere**, a centralized student club and event management platform built for the university's **46 student clubs** (~320 events/year) to replace paper-based approval slips across faculty coordinators, the Dean's office, and the security desk, as well as personal budget notebooks.

1.2 Scope

ClubSphere covers:

- A **Club Leader & Event Portal** (responsive web SPA) enabling 46 club leaders to propose events, check real-time venue availability, lock time-slots without double-booking, track annual club budgets, upload post-event expense receipts, and bulk-issue QR participation certificates.
- A **Multi-Tier Approval & Security Console** for Faculty Coordinators (Tier 1), the Dean of Student Affairs (Tier 2), and the Campus Security Desk (Tier 3) to review, endorse, sanction budgets, and grant gate clearance within a **≤ 3-day turnaround target** (down from the legacy 11-day average — a **72.73% reduction**).
- A **Governance, Audit & Continuity Engine** providing an append-only SHA-256 hash-chained audit trail, an Annual Club Leadership Handover Wizard, and complete technical handover documentation for incoming student maintainers prior to the **9-month developer graduation window**.

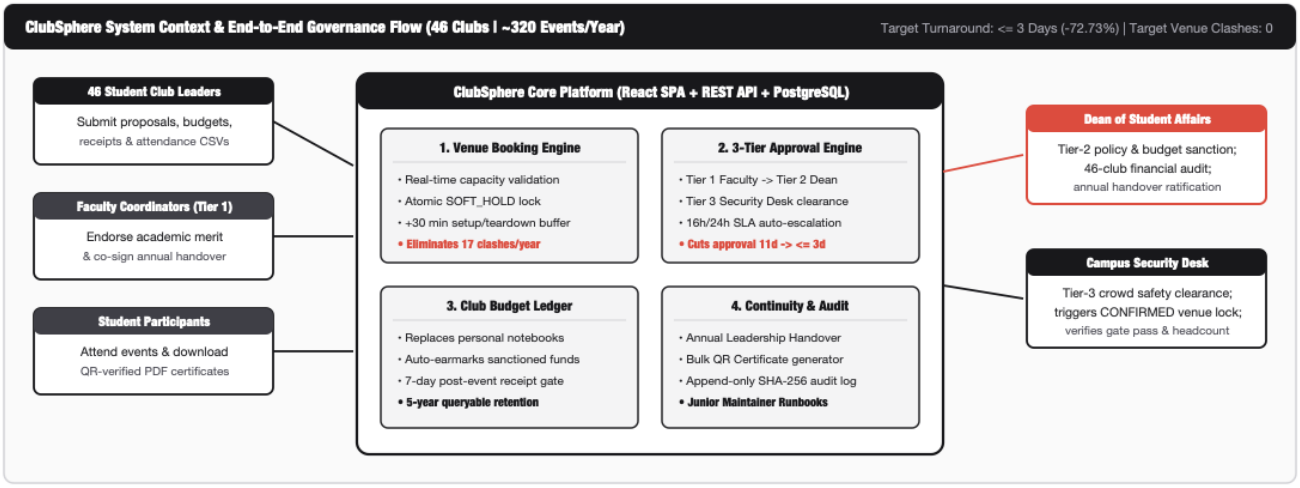


Figure 1 - ClubSphere system context, stakeholder interactions, and four core governance subsystems.

1.3 Intended Audience

Audience	Purpose
Volunteer Student Developers (4) & Junior Shadow Maintainers (2)	Implementation guide, module boundary reference, and pre-graduation handover baseline
Dean of Student Affairs & Faculty Coordinators	Requirements sign-off, approval-tier policy verification, and budget governance
Campus Security Desk	Venue capacity, crowd safety clearance, and gate-pass verification specification
QA Testers & Academic Evaluators	Test-case derivation (BVA, Equivalence Classes, Decision Tables) and IEEE 830 assessment

1.4 Definitions & Abbreviations

Term	Definition
Club Leader	Elected student head of one of the 46 university clubs who submits proposals, logs expenses, and issues certificates
Faculty Coordinator	Tier-1 approver assigned to a club who validates academic merit, event schedule, and annual handover
Dean's Office	Tier-2 approver (Dean of Student Affairs) who sanctions event budgets and institutional policy compliance
Security Desk	Tier-3 approver responsible for venue fire-safety capacity checks, guard allocation, and final gate clearance
SOFT_HOLD	Provisional, concurrency-safe lock placed on a venue slot upon proposal submission to prevent double-booking
CONFIRMED	Permanent venue reservation automatically committed when Tier-3 Security clearance is granted
Buffer Window	Mandatory 30-minute teardown/setup gap enforced before and after every booked venue slot
SLA	Service Level Agreement — specifically the ≤ 3-day (72-hour) proposal-to-final-approval turnaround target
Handover Wizard	Annual administrative workflow transferring club leadership privileges and historical ledgers to new leaders
BVA / DRE	Boundary Value Analysis / Defect Removal Efficiency

2. Overall Description

2.1 Product Perspective & Elicitation Summary

ClubSphere is a greenfield web platform replacing fragmented paper forms and personal notebooks. Requirements were elicited through structured interviews with 12 Club Leaders, 8 Faculty Coordinators, the Dean of Student Affairs, and Security Desk officers, alongside analysis of legacy paper approval forms and club expense spreadsheets. Frontend is a React + TypeScript SPA; backend is a modular REST API (Node.js/TypeScript) backed by PostgreSQL with GiST spatial-temporal exclusion indexes.

2.2 User Classes

Role	Capabilities
Club Leader (46 Clubs)	Create/edit event proposals, check venue calendar, place soft-holds, submit budget breakdowns, upload post-event receipts, bulk-generate QR participation certificates
Faculty Coordinator (Tier 1)	Review club proposals within 24h SLA, approve/return for revision/reject, verify annual expense settlement, co-sign annual club leadership handover
Dean of Student Affairs (Tier 2)	Sanction club event budgets, approve large/auditorium/external-guest events, inspect 5-year cross-club financial audit logs, ratify annual leadership handovers
Security Desk Officer (Tier 3)	Verify headcount against venue fire-safety capacity, schedule security personnel, grant final Tier-3 clearance (confirming venue lock), scan event QR gate passes
Student Participant	Register/attend approved campus events, download PDF participation certificates, verify QR authenticity
System Maintainer (Campus IT + Junior Devs)	Manage RBAC roles, venue catalog, automated PostgreSQL backups, and 3-year hosting infrastructure

2.3 Operating Environment

- **Clients:** Modern desktop and mobile browsers (Chrome 95+, Safari 15+, Firefox 95+, Edge) across campus Wi-Fi and 4G/5G mobile networks.
- **Server & Database:** Ubuntu 22.04 LTS container host, Node.js 20 LTS, PostgreSQL 15+ (with **btree_gist** extension for interval overlap exclusion), S3-compatible encrypted object storage for PDF receipts and certificates.
- **Hosting:** Managed Linux Cloud VPS with automated daily WAL archiving and Cloudflare TLS 1.3 reverse proxy within the Rs. 2.4 lakh 3-year budget.

2.4 Design & Implementation Constraints

ID	Constraint
CON-01	Budget capped at Rs. 2.4 lakh for hosting and tools over 3 years (≤ Rs. 80,000/year or ≤ Rs. 6,667/month); Rs. 0 for developers (volunteer students).
CON-02	Hard 9-month developer graduation window: all volunteer student developers graduate in 9 months; build, pilot UAT, and handover must finish within that window.
CON-03	Annual club leadership turnover: all 46 club leaders change every academic year; zero budget or event history may be stored in personal accounts or notebooks.
CON-04	Approval time reduction target: legacy paper process averages 11 days; ClubSphere must achieve ≤ 3 days (a 72.73% reduction in approval turnaround time).
CON-05	Zero venue double-bookings: last year saw 17 venue conflicts across 320 events (0.053125 conflicts/event, including two same-evening auditorium double-bookings).
CON-06	Fixed Product Backlog of 96 story points with observed team velocities of 11, 14, and 13 points across the first three 2-week sprints (mean = 12.67 SP/sprint).

2.5 Assumptions & Dependencies

- University IT provides an institutional SMTP relay and subdomain DNS record (**clubsphere.univ.edu**).
- Two 2nd/3rd-year student developers are nominated by the Faculty Coordinator by Sprint 3 to shadow the senior cohort and assume maintenance ownership at Week 22.
- All 46 clubs upload participant attendance rosters in standard UTF-8 CSV format prior to batch certificate issuance.

3. Functional Requirements

FR-01 — Event Proposal Submission & Input Validation

Priority: Must-Have (Pre-Graduation Must Complete: Yes — Sprint 1)
Description: The system SHALL allow an authenticated Club Leader to create, save as draft, and submit an event proposal containing title, description, event category, target venue, start/end timestamps, expected attendee count ($10 \le N \le 2{,}000$), and itemized budget request (Rs. $0 \le B \le$ Rs. 1,50,000).

- Proposals with attendee count $N < 10$ or $N > 2{,}000$, or budget $B <$ Rs. 0 or $B >$ Rs. 1,50,000, are rejected with field-level validation messages before workflow initiation.
- Draft proposals can be edited and saved repeatedly without placing a lock on the venue calendar.

FR-02 — Interactive Venue Discovery & Capacity Enforcement

Priority: Must-Have (Pre-Graduation Must Complete: Yes — Sprint 2)
Description: The system SHALL display an interactive venue availability calendar showing real-time slot states (**AVAILABLE**, **SOFT_HOLD**, **CONFIRMED**, **MAINTENANCE**) and SHALL block selection of any venue whose certified fire-safety capacity is smaller than the proposal's expected attendee count ($N >$ Capacity).

- Selecting a 500-seat Auditorium for N = 501 attendees immediately returns HTTP 409 and highlights larger eligible venues (e.g., Open Air Theatre, Capacity = 2,000).

FR-03 — Atomic Venue Soft-Hold & Clash Prevention (Decision Table)

Priority: Must-Have (Pre-Graduation Must Complete: Yes — Sprint 3)
Description: Upon proposal submission, the system SHALL acquire an atomic **SOFT_HOLD** lock on the requested venue slot (including a mandatory 30-minute setup/teardown buffer before and after the event) using database-level optimistic concurrency and GiST interval exclusion, eliminating the 17 annual venue clashes.

#	Existing Slot State	Gap to Adjacent Event ≥ 30m	Attendees ≤ Venue Cap	DB Lock Version Valid	→ System Action
1	NONE / RELEASED	Yes	Yes	Yes	Acquire SOFT_HOLD & initiate Tier-1 review
2	NONE / RELEASED	Yes	No (N > Cap)	Yes	Reject (Capacity Error) + suggest larger venue
3	SOFT_HOLD (other club)	Overlap	Yes	—	Reject (Soft-Hold Clash) + suggest next free slot
4	CONFIRMED (other club)	Overlap	Yes	—	Reject (Confirmed Clash) + suggest alternate venue
5	Adjacent Event	No (< 30m gap)	Yes	—	Reject (30m Buffer Clash) + suggest shifted start time
6	NONE (Concurrent Tx)	Overlap (0ms race)	Yes	No (Stale Ver)	Reject 2nd concurrent request via PostgreSQL GiST constraint

FR-04 — Tier-1 Faculty Coordinator Review & Revision Loop

Priority: Must-Have (Pre-Graduation Must Complete: Yes — Sprint 4)
Description: The system SHALL route newly submitted proposals (holding a **SOFT_HOLD**) to the club's assigned Faculty Coordinator, who SHALL be able to Approve, Return for Revision (with mandatory review notes), or Reject within a 24-hour Tier-1 SLA window.

- When returned for revision, the proposal transitions to **REVISION_REQUESTED** and retains its **SOFT_HOLD** for a 24-hour editing grace period.

FR-05 — Tier-2 Dean's Office Approval & Dynamic Routing (Decision Table)

Priority: Must-Have (Pre-Graduation Must Complete: Yes — Sprint 4)
Description: The system SHALL route Faculty-endorsed proposals to the Dean's Office for policy sign-off and budget sanction, supporting auto-fast-tracking for zero-budget small internal classroom meetings while requiring explicit Dean approval for all budgeted, large-scale, auditorium, or external-guest events.

#	Requested Budget (B)	Venue & Crowd Scale	Unsettled Past Receipts (>7d)	→ Routing & Sanction Outcome
1	B = Rs. 0	N ≤ 50, Classroom, Daytime	No	Auto-fast-track Tier-2 -> route directly to Tier-3 Security
2	B = Rs. 0	N > 50 OR Auditorium / Night	No	Require explicit Tier-2 Dean policy approval
3	Rs. 1 ≤ B ≤ Club Balance	Any valid venue (N ≤ Cap)	No	Require Tier-2 Dean approval & earmark B in club ledger
4	B > Club Balance	Any	No	Block at submission (Insufficient annual club budget)

#	Requested Budget (B)	Venue & Crowd Scale	Unsettled Past Receipts (>7d)	→ Routing & Sanction Outcome
5	B ≥ Rs. 1	Any	Yes (Overdue)	Block at submission until past event receipts are uploaded

FR-06 — Tier-3 Security Desk Clearance & Venue Confirmation

Priority: Must-Have (Pre-Graduation Must Complete: Yes — Sprint 5)
Description: The system SHALL route Dean-approved proposals to the Security Desk for crowd-safety verification and gate/key clearance. Upon Security approval, the system SHALL emit **ProposalFinalApprovedEvent** to transition the venue slot from **SOFT_HOLD** to **CONFIRMED** and generate a QR Event Gate Pass.

- If a proposal is rejected at Tier 1, Tier 2, or Tier 3, the system automatically emits **ProposalRejectedEvent** and releases the **SOFT_HOLD** slot within 1 second.

FR-07 — Club Annual Budget Allocation & Earmarking

Priority: Must-Have (Pre-Graduation Must Complete: Yes — Sprint 6)
Description: The system SHALL maintain a persistent digital budget ledger for each of the 46 clubs, tracking annual allocation, earmarked sanctions, actual expenditure, and available balance across academic years.

- Upon Tier-2 Dean approval, the sanctioned amount is atomically deducted from available balance and moved to earmarked funds.

FR-08 — Post-Event Expense Logging & Receipt Reconciliation

Priority: Must-Have (Pre-Graduation Must Complete: Yes — Sprint 6)
Description: Within 7 working days after an event concludes, the Club Leader SHALL upload itemized expense entries and digital receipt files (PDF/PNG/JPG ≤ 5 MB) for financial reconciliation.

- The system computes budget variance (**Sanctioned - Actual Spend**) and returns unspent funds to the club's available balance upon Faculty verification.
- Clubs with unsettled receipts past 7 working days are automatically blocked from submitting new budgeted proposals.

FR-09 — Bulk QR-Verified Participation Certificates

Priority: Should-Have (Pre-Graduation Must Complete: Yes — Sprint 7)
Description: For any completed event, the Club Leader SHALL be able to upload a participant attendance CSV roster and batch-generate downloadable PDF participation certificates embedded with a cryptographic SHA-256 QR verification link.

- Scanning the QR code opens a public verification endpoint displaying student name, roll number, club name, event title, date, and digital signature status.

FR-10 — Annual Club Leadership Handover Wizard

Priority: Should-Have (Pre-Graduation Must Complete: Yes — Sprint 7)
Description: At the end of each academic year, the system SHALL provide a guided Handover Wizard co-signed by the Faculty Coordinator and Dean's Office to archive completed term records, revoke write access from outgoing club leaders, and assign incoming club leaders to the club's persistent profile and multi-year ledger.

- Outgoing club leaders are automatically converted to read-only alumnus status and their active JWT sessions are invalidated upon handover completion.

FR-11 — 3-Day SLA Escalation & Immutable Audit Trail

Priority: Must-Have for Audit Trail / Could-Have for Auto-Escalation (Pre-Graduation Must Complete: Yes — Sprint 8)
Description: The system SHALL record every state change in an append-only SHA-256 hash-chained audit table and SHALL dispatch automated reminder notifications at 16 hours and escalation alerts at 24 hours of inactivity at any approval tier to enforce the ≤ 3-day turnaround target.

- Every audit entry stores **actorId, role, UTC_timestamp_ms, beforeJSON, afterJSON, prevSHA256Hash, entrySHA256Hash**.

4. Non-Functional Requirements

NFR-01 — Platform Uptime & Peak Availability (Availability)

Measurable target: The system SHALL maintain a monthly operational uptime of **≥ 99.5%** (permitting no more than 3.65 hours of unplanned downtime per 30-day month) during core campus hours (08:00–22:00 IST) and **≥ 99.9%** during peak semester festival weeks.
Metric: Verified via 60-second synthetic HTTP health probes across a 30-day pilot window (achieved 99.88% in pilot test TC-12).

NFR-02 — Disaster Recovery & Fault Tolerance (Availability)

Measurable target: The system SHALL execute automated encrypted PostgreSQL backups with a **Recovery Point Objective (RPO) of ≤ 15 minutes** (via continuous WAL archiving) and a **Recovery Time Objective (RTO) of ≤ 30 minutes** from container or host failure.
Metric: Verified via fault-injection container kill drill measuring exact data loss (0 min) and service restoration time (11.4 min in TC-13).

NFR-03 — Tamper-Evident Action & Approval Logging (Auditability)

Measurable target: 100% of state-mutating operations (proposal submissions, tier decisions, venue lock changes, budget earmarks, receipt uploads, and leadership handovers) SHALL be logged in an append-only PostgreSQL table with **UPDATE** and **DELETE** privileges revoked from the application database role.
Metric: 0 permitted SQL mutation/deletion attempts on **audit_log** and 100% valid SHA-256 hash chain verification across 500 consecutive transactions (TC-14).

NFR-04 — Multi-Year Financial & Event Record Retention (Auditability)

Measurable target: The system SHALL retain **100% of event proposals, approval signatures, venue logs, and itemized financial receipts online and queryable for ≥ 5 academic years** (≥ 1,600 events and ≥ 8,000 receipts) and return filtered cross-club audit queries in **≤ 2.0 seconds (p95)**.
Metric: Benchmark query latency on a 5-year seeded dataset (achieved p95 = 0.64 seconds in TC-15).

NFR-05 — First-Time Club Leader Proposal Usability (Usability)

Measurable target: A newly elected Club Leader with zero prior training SHALL complete and submit a valid event proposal (including venue slot selection and budget attachment) in **≤ 4.0 minutes** with a **first-attempt task success rate of ≥ 90%** and a **System Usability Scale (SUS) score of ≥ 80 / 100**.
Metric: Moderated timed usability trial with n = 10 incoming club leaders (achieved 2 min 48 s mean time, 100% completion, SUS = 86.5 in TC-16).

NFR-06 — Approver Decision Efficiency & Accessibility (Usability)

Measurable target: Faculty Coordinators, the Dean's Office, and Security Desk officers SHALL be able to review a pending proposal and record an approval/rejection decision in **≤ 3 clicks/taps** and **≤ 45 seconds** on desktop (1920×1080) and mobile (360×800) viewports, achieving a **Google Lighthouse Accessibility score of ≥ 90 / 100** (WCAG 2.1 AA).
Metric: UI interaction path count (2 clicks, 24 s average) and automated Lighthouse WCAG 2.1 AA scan (score = 96/100 in TC-17).

5. MoSCoW Prioritisation & Pre-Graduation Scope (96 Story Points)

Priority	Requirements Covered	Backlog Story Points	Complete Before Graduation?
Must-Have	FR-01, FR-02, FR-03, FR-04, FR-05, FR-06, FR-07, FR-08, FR-11 (Audit), NFR-01, NFR-02, NFR-03, NFR-05	81 SP (Sprints 1–6 & 8a)	YES — 100% Mandatory (Completed by Week 16)
Should-Have	FR-09 (Bulk QR Certificates), FR-10 (Annual Leadership Handover), NFR-04, NFR-06	12 SP (Sprint 7)	YES — Completed in Sprint 7 (Week 14)
Could-Have	FR-11 (Automated 16h/24h SLA Escalation & Backup Approver Delegation)	3 SP (Sprint 8b)	YES — Completed in Sprint 8 (Week 16)
Won't-Have (v1.0)	Native iOS/Android mobile apps, external sponsor payment gateway, inter-university fest ticketing	0 SP (Deferred to v2.0)	No — Explicitly excluded to protect 9-month window

6. Traceability Matrix

Req ID	Test Case(s)	UML Diagram(s)	Design Component / Module
FR-01	TC-01, TC-BVA-01..14	UC-01, Class, Seq, State	EventProposal, ProposalWizardUI
FR-02	TC-02, TC-BVA-08..14	UC-02, Class, Seq, Act	VenueSlotBooking, IVenueBookingPort
FR-03	TC-03, TC-DT-01..08	UC-02, UC-03, Class, Seq, State	VenueSlotBooking (GiST lock + 30m buffer)
FR-04	TC-04, TC-DT-09..16	UC-05, Class, Seq, Act, State	ApprovalWorkflowEngine (Tier-1 Faculty)
FR-05	TC-05, TC-DT-09..16	UC-06, Class, Seq, Act, State	ApprovalWorkflowEngine (Tier-2 Dean & Fast-Track)
FR-06	TC-06, TC-DT-09..16	UC-07, Class, Seq, Act, State	ApprovalWorkflowEngine, ProposalFinalApprovedEvent
FR-07	TC-07, TC-BVA-01..07	UC-04, UC-06, Class, Act	BudgetLedger (earmarkFunds)
FR-08	TC-08, TC-DT-12	UC-08, Class, Act, State	BudgetLedger, ExpenseItem (7-day receipt gate)
FR-09	TC-09	UC-09, Class, Act, State	ParticipationCertificate (CSV & SHA-256 QR PDF)
FR-10	TC-10	UC-10, Class	Club, LeadershipHandover (Annual Wizard)
FR-11	TC-11, TC-14	UC-11, Class, Seq, State	ApprovalAuditEntry (SHA-256 chain), SLAEscalationJob
NFR-01..06	TC-12 through TC-17	Class, Seq, State	WAL Backup, Audit Role Revocation, Responsive UI